

# Digital Electronic Systems Design Project Report

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## Timing Analysis

From our implementation, the timing analysis yields compliant results for both setup and hold constraints at the target operating frequency of 150 MHz.

Regarding the setup time, the most critical path is located within the output selector (Path 21). While the design requirement demands a 150 MHz clock, this implementation achieves a positive slack of 0.230 ns. This safety margin allows the circuit to potentially operate at a maximum frequency of approximately 160 MHz.

Concerning the hold time, the critical path is identified within the `axis_fifo_0` module (Path 31). The required hold time constraint is 0 ns, and our implementation guarantees a safe positive hold slack of 0.230 ns, ensuring that no race conditions occur.

The detailed parameters resulting from the implementation are summarized in Table 1 for setup time and Table 2 for hold time below.

Tabel 1: Timing Summary: Setup Time (Path 21)

| Parameter          | Value  | Unit |
|--------------------|--------|------|
| Slack              | 0.230  | ns   |
| Requirement        | 6.627  | ns   |
| Data Path Delay    | 6.235  | ns   |
| <i>Logic Delay</i> | 3.288  | ns   |
| <i>Route Delay</i> | 2.947  | ns   |
| Clock Path Skew    | -0.113 | ns   |

Tabel 2: Timing Summary: Hold Time (Path 31)

| Parameter          | Value | Unit |
|--------------------|-------|------|
| Slack              | 0.049 | ns   |
| Requirement        | 0.000 | ns   |
| Data Path Delay    | 0.533 | ns   |
| <i>Logic Delay</i> | 0.184 | ns   |
| <i>Route Delay</i> | 0.349 | ns   |
| Clock Path Skew    | 0.352 | ns   |

## Power Analysis

At the operating frequency of 150 MHz, the following implementation's power consumption is as showed in the following tabels.

Tabel 3: Power Analysis Summary.

| Parameter                 | Value | Unit |
|---------------------------|-------|------|
| Total On-Chip Power       | 0.335 | W    |
| Design Power Budget       | N/A   | –    |
| Power Budget Margin       | N/A   | –    |
| Junction Temperature      | 26.7  | °C   |
| Thermal Margin            | 58.3  | °C   |
| Effective $\theta_{JA}$   | 5.0   | °C/W |
| Power to Off-Chip Devices | 0     | W    |
| Confidence Level          | Low   | –    |

Tabel 4: On-Chip Power Breakdown.

| Component            | Power        | Unit     | Ratio (%)  |
|----------------------|--------------|----------|------------|
| <b>Dynamic</b>       | <b>0.252</b> | <b>W</b> | <b>78%</b> |
| <i>MMCM</i>          | 0.126        | W        | 48%        |
| <i>BRAM</i>          | 0.078        | W        | 30%        |
| <i>Clocks</i>        | 0.018        | W        | 7%         |
| <i>I/O</i>           | 0.017        | W        | 6%         |
| <i>Signals</i>       | 0.012        | W        | 5%         |
| <i>Logic</i>         | 0.009        | W        | 3%         |
| <i>DSP</i>           | 0.002        | W        | 1%         |
| <b>Device Static</b> | <b>0.074</b> | <b>W</b> | <b>22%</b> |